

Intelligent Warehouse Management System

with ML-Powered Demand Forecasting

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MAY 2026

ABSTRACT

A full-stack, role-based Warehouse Management System for multi-location retail operations with an XGBoost ML demand forecasting engine. Replaces manual spreadsheet-based tracking with real-time digital workflows across three dedicated role dashboards — Owner, Warehouse Staff, Branch Staff — enforced by PostgreSQL Row Level Security. Vectorized ML predictions outperform the Weighted Moving Average baseline with graceful offline fallback.

01 INTRODUCTION

Multi-branch retail businesses rely on spreadsheets and manual processes, causing stock discrepancies, data loss, and delayed decisions. Existing systems lack role-specific access and use naive forecasting — leading to simultaneous overstock and stockouts.

02 PROJECT OBJECTIVES

- Role-Based Access**
Dedicated dashboards for Owner, Warehouse & Branch Staff with RLS-enforced data boundaries.
- Full Inventory Lifecycle**
Intake, transfers, POS, loans, returns, corrections & auditing in one unified system.
- ML Demand Forecasting**
XGBoost model outperforming the WMA baseline with per-product, per-location predictions.
- Real-Time & Multilingual**
Live Supabase updates with EN, RU, UZ-Latin & UZ-Cyrillic interfaces.

03 METHODOLOGY

PHASE 1 — DESIGN & DATABASE

Field analysis of retail workflows. PostgreSQL schema with RLS, Google OAuth & atomic stored procedures to prevent race conditions.

PHASE 2 — CORE SYSTEM & DASHBOARDS

Full transaction lifecycle, inter-branch approval workflows, three role dashboards, analytics, auditing & four-language support.

PHASE 3 — ML INTEGRATION

XGBoost trained on 26 engineered features, served via FastAPI with vectorized batch inference & WMA/ML toggle.

PHASE 4 — ADVANCED FEATURES

Inventory audit system, Smart Restock engine, real-time analytics with branch comparison charts & automated Excel report generation.

04 RESULTS

KEY ACHIEVEMENT

A reactive, paper-based operation becomes a proactive, data-driven business — real-time inventory control and ML-guided restocking, available to every role, in every language.

05 PROJECT OUTPUT

WEB APPLICATION

Production-ready SPA with three role dashboards — Owner, Warehouse Staff, Branch Staff — secured by PostgreSQL Row Level Security and deployed on Vercel.

ML PREDICTION SERVER

FastAPI + XGBoost inference engine with vectorized batch predictions and automatic WMA fallback when offline.

INVENTORY MANAGEMENT SYSTEM

Seven transaction types covering the full lifecycle: intake, sale, return, loan, loan return, transfer, and correction.

06 KEY CHALLENGES

RACE CONDITIONS IN STOCK OPERATIONS

Resolved with PostgreSQL stored procedures using `SELECT ... FOR UPDATE` row-level locking, ensuring atomic deduct & add operations under concurrent users.

SUPABASE 1,000-ROW API LIMIT

Built a custom `fetchAll()` utility that auto-paginates via `.range()`, fetching all records in batches and concatenating results seamlessly.

ML BATCH PREDICTION TIMEOUT

Refactored from per-product loops to vectorized single-call inference — reducing prediction time from 30+ seconds to ~3 seconds for all products.

07 CONCLUSION

This project delivers a full-stack, role-based WMS that replaces manual workflows at a real multi-branch retail business in Uzbekistan with a real-time digital platform.

Role-enforced access control secures financial data while the ML forecasting layer transforms restocking from reactive to proactive — reducing stockouts and overstock simultaneously.

CORE CONTRIBUTION

A unified platform covering the complete warehouse lifecycle, with role-based data security and intelligent demand forecasting — purpose-built for real multi-branch retail operations.

08 RECOMMENDATION & FUTURE WORK



Mobile Application

Native app with barcode & QR scanning for floor-level stock counts and sales operations.



Supplier Auto-Reordering

Auto-generate purchase orders when ML predicts stock below threshold — closing the prediction-to-procurement loop.



Advanced Analytics

Dead stock identification, ABC classification, seasonal heatmaps & scheduled email report delivery.

09 ACKNOWLEDGMENT

We sincerely thank our graduation project supervisor for their continuous guidance and support throughout the development process.

Special thanks to the retail business that served as our pilot site, and to the open-source communities behind React, Supabase, XGBoost, and FastAPI.